

PRAYCG Master Comprehensive Analysis Suite

Elaborated module explanation report for v1.5.5

Open-source methods handout - module behavior, outputs, interpretation, and claim boundaries

Core orientation
The Master Comprehensive Suite is a research workbench, not a single test. The recommended public/confirmatory path is boxed: Timing/QC + StimulusFingerprint/CET-R + artifact/confound gates -> A-MRED and MRED-Peak/Resolution. Everything else should be treated as a secondary summary or exploratory convergence layer unless explicitly preregistered as an endpoint.

Executive map

Question	Primary module family	Why this matters
Did the run happen cleanly?	Run metadata, stream inventory, ALS, channel map, artifact/confound gates	No analysis module can rescue a run with broken timing, missing streams, or dominant artifact.
Did the Target event pass?	A-MRED / MRED-Peak	This is the clearest current endpoint for acute meaning-recognition plus delayed integration.
Was the response more like closure than shock?	MRED-Resolution / NUPI / Baseline2 / EET	Some stimuli may produce reflective recovery rather than a sharp peak.
Could it just be the movie/cues?	StimulusFingerprint / CET / CET-R	These modules test whether low-level exogenous structure explains the effect.
Could it just be task burden?	OCM025 / RSM / CVB / SquintProxy / TTI	The Override condition must be modeled as its own cognitive machine.
How should a non-expert read the output?	Offline Interpretive Report Generator	Produces deterministic local explanation without requiring internet access or an AI assistant.

Module families and how they connect

The suite should be read as a chain. MediaPrep defines the external stimulus; the runner records what was loaded and what the participant reported; the Master Suite builds the feature table; the primary endpoint modules compress that table into A-MRED and MRED-Peak/Resolution; secondary modules summarize immersion, extraction, polarity, and afterstate; exploratory modules test convergence or failure modes. A positive interpretation requires convergence across the chain, not a single attractive graph.

Do not do this
Do not interpret a gamma spike alone as meaning. Do not interpret a blink as memory encoding. Do not interpret EET as replay. Do not interpret NUPI as thermodynamic heat. Do not interpret self-report as proof. Each module has a narrow job.

1. Run metadata and media identity

Tier: Primary QC

Verifies participant/session labels, protocol version, branch videos, cue schedule, anchor file, hash identity, and configured durations.

What the module is trying to prevent

This module exists primarily to prevent a false-positive interpretation. It asks whether the run, timing, stimulus, or signal quality is strong enough to allow any downstream claim. If this layer fails, the correct output is not a weaker positive result; it is a downgraded or non-interpretable run.

Inputs and assumptions

Runner event log, run_config_media_selection.json, media hashes, cue schedule summary.

The assumptions are narrow: the input files must correspond to the same run, use the same branch timing, and preserve the condition labels expected by the suite. If the module is anchor-based, anchor IDs and rendered times must match the final rendered stimulus rather than a remembered source-video time.

Algorithmic logic

Parse setup markers and sidecar JSON. Confirm Target/Override hash match. Record cue count, expected sum, baseline/washout durations, anchor-file status, channel-map status, and PRAYCG version.

In practice, the module reduces high-dimensional time series into branch-level, anchor-level, or cue-level summaries. The exact weights and thresholds are intended to be transparent and modifiable rather than hidden. For public replication, the lock-in file should freeze thresholds before analysis.

Outputs and where to find them

run_identity_summary.csv, module registry rows, interpretation JSON fields.

These outputs usually live in the Master Suite output folder under tables/. Visualizer-specific sidecars are separate and should not be confused with canonical analysis tables.

How to interpret it

A valid run should show the selected media and cue schedule unambiguously. Target and Override should be hash-identical when the task differs only by instruction.

The safest interpretation always asks three questions: Did Target exceed Control? Did Target exceed Override? Did artifact, timing, stimulus drive, or self-report contradict the interpretation?

How it can fail

Missing config, Target/Override mismatch, unrecorded anchor file, absent cue schedule, or unknown protocol version downgrades claims.

Safe public wording

Recommended wording

This module supports the analysis chain. It should be read as a bounded measurement or QC layer, not a standalone claim.

2. Stream inventory and timebase reconstruction

Tier: Primary QC

Determines which LSL streams are present and whether physiology can be aligned to event markers.

What the module is trying to prevent

This module exists primarily to prevent a false-positive interpretation. It asks whether the run, timing, stimulus, or signal quality is strong enough to allow any downstream claim. If this layer fails, the correct output is not a weaker positive result; it is a downgraded or non-interpretable run.

Inputs and assumptions

XDF streams, StasisMarkers, OpenBCIStatusMarkers, sample timestamps.

The assumptions are narrow: the input files must correspond to the same run, use the same branch timing, and preserve the condition labels expected by the suite. If the module is anchor-based, anchor IDs and rendered times must match the final rendered stimulus rather than a remembered source-video time.

Algorithmic logic

Inventory stream names, sample rates, channel counts, durations, time gaps, and marker intervals. When raw timestamps are unreliable, reconstruct the OpenBCI timebase using sample-index and status markers.

In practice, the module reduces high-dimensional time series into branch-level, anchor-level, or cue-level summaries. The exact weights and thresholds are intended to be transparent and modifiable rather than hidden. For public replication, the lock-in file should freeze thresholds before analysis.

Outputs and where to find them

stream_inventory_corrected.csv, phase_coverage.csv.

These outputs usually live in the Master Suite output folder under tables/. Visualizer-specific sidecars are separate and should not be confused with canonical analysis tables.

How to interpret it

Successful reconstruction supports phase-sliced features even when raw XDF timestamps are imperfect.

The safest interpretation always asks three questions: Did Target exceed Control? Did Target exceed Override? Did artifact, timing, stimulus drive, or self-report contradict the interpretation?

How it can fail

Missing EEG stream, major dropout, unsolved timebase inconsistency, or poor phase coverage prevents strong interpretation.

Safe public wording

Recommended wording
This module supports the analysis chain. It should be read as a bounded measurement or QC layer, not a

standalone claim.

3. ALS/PT19 display-timing QC

Tier: Primary QC

Checks whether physical screen transitions were detected at Control, Target, and Override starts.

What the module is trying to prevent

This module exists primarily to prevent a false-positive interpretation. It asks whether the run, timing, stimulus, or signal quality is strong enough to allow any downstream claim. If this layer fails, the correct output is not a weaker positive result; it is a downgraded or non-interpretable run.

Inputs and assumptions

ALS_PT19_Timing stream or OpenBCI analog AUX, branch start markers, cue/media timing manifest.

The assumptions are narrow: the input files must correspond to the same run, use the same branch timing, and preserve the condition labels expected by the suite. If the module is anchor-based, anchor IDs and rendered times must match the final rendered stimulus rather than a remembered source-video time.

Algorithmic logic

Detect black-white-black or timing-square pulses around branch-start markers. Estimate onset offset, duration, clipping, and pulse presence.

In practice, the module reduces high-dimensional time series into branch-level, anchor-level, or cue-level summaries. The exact weights and thresholds are intended to be transparent and modifiable rather than hidden. For public replication, the lock-in file should freeze thresholds before analysis.

Outputs and where to find them

als_start_pulse_qc.csv, ALS timing summary.

These outputs usually live in the Master Suite output folder under tables/. Visualizer-specific sidecars are separate and should not be confused with canonical analysis tables.

How to interpret it

A clean pass strengthens display-onset alignment. ALS belongs to external input $u(t)$, not biology.

The safest interpretation always asks three questions: Did Target exceed Control? Did Target exceed Override? Did artifact, timing, stimulus drive, or self-report contradict the interpretation?

How it can fail

Missed or ambiguous pulses downgrade anchor timing and endpoint claim grade.

Safe public wording

Recommended wording

This module supports the analysis chain. It should be read as a bounded measurement or QC layer, not a
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standalone claim.

4. Channel map and ROI audit

Tier: Primary QC

Checks whether channel-to-electrode mapping is confirmed before ROI-based interpretation.

What the module is trying to prevent

This module exists primarily to prevent a false-positive interpretation. It asks whether the run, timing, stimulus, or signal quality is strong enough to allow any downstream claim. If this layer fails, the correct output is not a weaker positive result; it is a downgraded or non-interpretable run.

Inputs and assumptions

channel_map.csv, event markers, run configuration.

The assumptions are narrow: the input files must correspond to the same run, use the same branch timing, and preserve the condition labels expected by the suite. If the module is anchor-based, anchor IDs and rendered times must match the final rendered stimulus rather than a remembered source-video time.

Algorithmic logic

Parse PRAYCG16 channel map markers and confidence flags. Mark ROI use as allowed, cautioned, or blocked.

In practice, the module reduces high-dimensional time series into branch-level, anchor-level, or cue-level summaries. The exact weights and thresholds are intended to be transparent and modifiable rather than hidden. For public replication, the lock-in file should freeze thresholds before analysis.

Outputs and where to find them

channel_map_summary.csv or module registry field.

These outputs usually live in the Master Suite output folder under tables/. Visualizer-specific sidecars are separate and should not be confused with canonical analysis tables.

How to interpret it

LOCKED channel map allows frontal/parietal/occipital/temporal ROI labels to be used cautiously.

The safest interpretation always asks three questions: Did Target exceed Control? Did Target exceed Override? Did artifact, timing, stimulus drive, or self-report contradict the interpretation?

How it can fail

Unconfirmed channel map blocks ROI-specific claims even if global features can still be analyzed.

Safe public wording

Recommended wording

This module supports the analysis chain. It should be read as a bounded measurement or QC layer, not a

standalone claim.

5. Feature table generation

Tier: Primary engine

Creates the time-resolved table used by all downstream modules.

What the module is trying to prevent

This module supports the core analysis chain. It should be interpreted only in relation to timing, artifact, Control, Override, and stimulus-regressor checks.

Inputs and assumptions

Cleaned/filtered EEG, event markers, HR/RR, respiration, ALS, artifact channels, StimulusFingerprint regressors where available.

The assumptions are narrow: the input files must correspond to the same run, use the same branch timing, and preserve the condition labels expected by the suite. If the module is anchor-based, anchor IDs and rendered times must match the final rendered stimulus rather than a remembered source-video time.

Algorithmic logic

Slice by phase, compute sliding-window features, spectral bands, ROI summaries, artifact sentinels, autonomic summaries, and condition labels. Typically writes 1 s rows; OCM025 uses 0.25 s cue windows.

In practice, the module reduces high-dimensional time series into branch-level, anchor-level, or cue-level summaries. The exact weights and thresholds are intended to be transparent and modifiable rather than hidden. For public replication, the lock-in file should freeze thresholds before analysis.

Outputs and where to find them

*_time_resolved_feature_frame.csv.

These outputs usually live in the Master Suite output folder under tables/. Visualizer-specific sidecars are separate and should not be confused with canonical analysis tables.

How to interpret it

This is the canonical physiology feature table. Visualizer *_features_used.csv is only the subset used for a render.

The safest interpretation always asks three questions: Did Target exceed Control? Did Target exceed Override? Did artifact, timing, stimulus drive, or self-report contradict the interpretation?

How it can fail

Missing feature table means most downstream modules cannot run.

Safe public wording

Recommended wording

This module supports the analysis chain. It should be read as a bounded measurement or QC layer, not a standalone claim.

6. ArtifactScore

Tier: Primary gate

Quantifies windows where EEG features may be contaminated by blink, jaw, motion, flatline, or high-frequency noise.

What the module is trying to prevent

This module exists primarily to prevent a false-positive interpretation. It asks whether the run, timing, stimulus, or signal quality is strong enough to allow any downstream claim. If this layer fails, the correct output is not a weaker positive result; it is a downgraded or non-interpretable run.

Inputs and assumptions

EEG channels, Fp1/Fp2/frontal channels, high-frequency sentinel bands, peak-to-peak amplitude.

The assumptions are narrow: the input files must correspond to the same run, use the same branch timing, and preserve the condition labels expected by the suite. If the module is anchor-based, anchor IDs and rendered times must match the final rendered stimulus rather than a remembered source-video time.

Algorithmic logic

Combine normalized peak-to-peak, line/high-frequency sentinel activity, ocular/frontal outlier features, and dropout/flatline checks. Penalize or veto contaminated windows.

In practice, the module reduces high-dimensional time series into branch-level, anchor-level, or cue-level summaries. The exact weights and thresholds are intended to be transparent and modifiable rather than hidden. For public replication, the lock-in file should freeze thresholds before analysis.

Outputs and where to find them

artifact_score columns, artifact tables, module veto flags.

These outputs usually live in the Master Suite output folder under tables/. Visualizer-specific sidecars are separate and should not be confused with canonical analysis tables.

How to interpret it

ArtifactScore has veto power. A gamma/theta event that is artifact-driven is not an endpoint.

The safest interpretation always asks three questions: Did Target exceed Control? Did Target exceed Override? Did artifact, timing, stimulus drive, or self-report contradict the interpretation?

How it can fail

High artifact in candidate windows converts biological interpretation into artifact/confound caution.

Safe public wording

Recommended wording

This module supports the analysis chain. It should be read as a bounded measurement or QC layer, not a standalone claim.

7. StimulusFingerprint v1.8

Tier: Primary guardrail

Extracts exogenous stimulus regressors from the final rendered media.

What the module is trying to prevent

This module exists primarily to prevent a false-positive interpretation. It asks whether the run, timing, stimulus, or signal quality is strong enough to allow any downstream claim. If this layer fails, the correct output is not a weaker positive result; it is a downgraded or non-interpretable run.

Inputs and assumptions

Control/Target/Override MP4s, cue schedule JSON, locked anchor JSON if available.

The assumptions are narrow: the input files must correspond to the same run, use the same branch timing, and preserve the condition labels expected by the suite. If the module is anchor-based, anchor IDs and rendered times must match the final rendered stimulus rather than a remembered source-video time.

Algorithmic logic

Compute luminance, visual change, histogram delta, cut events, flash proxy, audio RMS/envelope derivative, cue trains, cue phase, ALS/start-pulse regressors, and anchor stimulus vectors.

In practice, the module reduces high-dimensional time series into branch-level, anchor-level, or cue-level summaries. The exact weights and thresholds are intended to be transparent and modifiable rather than hidden. For public replication, the lock-in file should freeze thresholds before analysis.

Outputs and where to find them

qc/stimulus_exogenous_regressor_frame_all_conditions.csv, cet_regressors_all_conditions.csv, stimulus_rhythm_summary_all_conditions.csv.

These outputs usually live in the Master Suite output folder under tables/. Visualizer-specific sidecars are separate and should not be confused with canonical analysis tables.

How to interpret it

This lets CET-R test whether physiology is explained by stimulus rhythm, cue rhythm, luminance, cuts, motion, or audio envelope.

The safest interpretation always asks three questions: Did Target exceed Control? Did Target exceed Override? Did artifact, timing, stimulus drive, or self-report contradict the interpretation?

How it can fail

Missing Target/Override regressors makes CET-R partial. Human QC is still required for control audio unintelligibility.

Safe public wording

Recommended wording

The stimulus-regressor check did or did not explain the observed pattern. This is a guardrail, not a proof of meaning.

8. CET - Cinematic Entrainment Tracking

Tier: Primary guardrail / exploratory

Estimates how much EEG/NIP-like features track external film/cue/audio/visual rhythm.

What the module is trying to prevent

This module exists primarily to prevent a false-positive interpretation. It asks whether the run, timing, stimulus, or signal quality is strong enough to allow any downstream claim. If this layer fails, the correct output is not a weaker positive result; it is a downgraded or non-interpretable run.

Inputs and assumptions

StimulusFingerprint regressors and feature table.

The assumptions are narrow: the input files must correspond to the same run, use the same branch timing, and preserve the condition labels expected by the suite. If the module is anchor-based, anchor IDs and rendered times must match the final rendered stimulus rather than a remembered source-video time.

Algorithmic logic

Fit stimulus-side regressors to physiology features within each branch. Evaluate train and blocked-cross-validation R2.

In practice, the module reduces high-dimensional time series into branch-level, anchor-level, or cue-level summaries. The exact weights and thresholds are intended to be transparent and modifiable rather than hidden. For public replication, the lock-in file should freeze thresholds before analysis.

Outputs and where to find them

cet_residualization_model_summary.csv, cet_cue_tracking_summary.csv.

These outputs usually live in the Master Suite output folder under tables/. Visualizer-specific sidecars are separate and should not be confused with canonical analysis tables.

How to interpret it

High generalizable R2 means apparent effects may be stimulus tracking. Low/negative blocked-CV suggests stimulus regressors do not explain the module pattern.

The safest interpretation always asks three questions: Did Target exceed Control? Did Target exceed Override? Did artifact, timing, stimulus drive, or self-report contradict the interpretation?

How it can fail

Weak regressors, missing audio/video features, or overfit models prevent strong conclusions.

Safe public wording

Recommended wording
The stimulus-regressor check did or did not explain the observed pattern. This is a guardrail, not a proof of meaning.

9. CET-R - Residualized meaning metrics

Tier: Primary guardrail

Re-tests NIP/MRED/TTI after regressing out external media rhythms.

What the module is trying to prevent

This module exists primarily to prevent a false-positive interpretation. It asks whether the run, timing, stimulus, or signal quality is strong enough to allow any downstream claim. If this layer fails, the correct output is not a weaker positive result; it is a downgraded or non-interpretable run.

Inputs and assumptions

Feature table, CII/NIP/MRED metrics, StimulusFingerprint regressors.

The assumptions are narrow: the input files must correspond to the same run, use the same branch timing, and preserve the condition labels expected by the suite. If the module is anchor-based, anchor IDs and rendered times must match the final rendered stimulus rather than a remembered source-video time.

Algorithmic logic

Model features as functions of exogenous regressors, compute residuals, then recompute anchor integrals/contrasts.

In practice, the module reduces high-dimensional time series into branch-level, anchor-level, or cue-level summaries. The exact weights and thresholds are intended to be transparent and modifiable rather than hidden. For public replication, the lock-in file should freeze thresholds before analysis.

Outputs and where to find them

cet_residualized_cii_anchor_integrals.csv, residualized summaries.

These outputs usually live in the Master Suite output folder under tables/. Visualizer-specific sidecars are separate and should not be confused with canonical analysis tables.

How to interpret it

If Target remains greater than Control/Override after CET-R, the result is less likely to be reducible to raw stimulus rhythm.

The safest interpretation always asks three questions: Did Target exceed Control? Did Target exceed Override? Did artifact, timing, stimulus drive, or self-report contradict the interpretation?

How it can fail

If effects vanish after residualization, meaning-specific interpretation weakens.

Safe public wording

Recommended wording
The stimulus-regressor check did or did not explain the observed pattern. This is a guardrail, not a proof of meaning.

10. GammaScalpel

Tier: Feature extraction / secondary

Separates lower-gamma candidate activity into narrower bands and context labels instead of treating gamma as meaning.

What the module is trying to prevent

This module supports the core analysis chain. It should be interpreted only in relation to timing, artifact, Control, Override, and stimulus-regressor checks.

Inputs and assumptions

EEG feature windows, ROI labels, artifact scores, condition labels.

The assumptions are narrow: the input files must correspond to the same run, use the same branch timing, and preserve the condition labels expected by the suite. If the module is anchor-based, anchor IDs and rendered times must match the final rendered stimulus rather than a remembered source-video time.

Algorithmic logic

Subdivide lower gamma, e.g. 30-35, 35-40, 40-45 Hz, compare Target/Control/Override, separate power from phase-locking, apply artifact and visual/task penalties.

In practice, the module reduces high-dimensional time series into branch-level, anchor-level, or cue-level summaries. The exact weights and thresholds are intended to be transparent and modifiable rather than hidden. For public replication, the lock-in file should freeze thresholds before analysis.

Outputs and where to find them

gammascalpel_microband_phase_summary.csv, MeaningGamma/TaskGamma components.

These outputs usually live in the Master Suite output folder under tables/. Visualizer-specific sidecars are separate and should not be confused with canonical analysis tables.

How to interpret it

Gamma is an artifact-sensitive candidate work-signal proxy, not a biomarker. It contributes to MR only if gates survive.

The safest interpretation always asks three questions: Did Target exceed Control? Did Target exceed Override? Did artifact, timing, stimulus drive, or self-report contradict the interpretation?

How it can fail

EMG/ocular/artifact, cue/task load, or visual features can explain gamma-like changes.

Safe public wording

Recommended wording
This module contributed an artifact-sensitive recognition/work-signal proxy. It is not a standalone meaning biomarker.

11. Temporal Semantic Proxy (TSP)

Tier: Feature extraction / secondary

A composite proxy for scene-timed semantic recognition pressure.

What the module is trying to prevent

This module supports the core analysis chain. It should be interpreted only in relation to timing, artifact, Control, Override, and stimulus-regressor checks.

Inputs and assumptions

MeaningGamma proxy, taper/peak features, artifact penalties, condition labels, anchor windows.

The assumptions are narrow: the input files must correspond to the same run, use the same branch timing, and preserve the condition labels expected by the suite. If the module is anchor-based, anchor IDs and rendered times must match the final rendered stimulus rather than a remembered source-video time.

Algorithmic logic

Score temporally localized recognition-like events using meaning-weighted gamma/TSP components, taper detection, and artifact-corrected normalization.

In practice, the module reduces high-dimensional time series into branch-level, anchor-level, or cue-level summaries. The exact weights and thresholds are intended to be transparent and modifiable rather than hidden. For public replication, the lock-in file should freeze thresholds before analysis.

Outputs and where to find them

TSP columns, TSP event candidates, visual overlays.

These outputs usually live in the Master Suite output folder under tables/. Visualizer-specific sidecars are separate and should not be confused with canonical analysis tables.

How to interpret it

TSP marks candidate semantic strike timing. It is not proof of comprehension or memory.

The safest interpretation always asks three questions: Did Target exceed Control? Did Target exceed Override? Did artifact, timing, stimulus drive, or self-report contradict the interpretation?

How it can fail

TSP can be driven by visual salience, audio, task, or artifact if not controlled.

Safe public wording

Recommended wording
This module contributed an artifact-sensitive recognition/work-signal proxy. It is not a standalone meaning biomarker.

12. CandidateLocal KHT-topo

Tier: Exploratory convergence

Tests local reciprocal coupling between observed work-signal features and inferred network-state topology proxies.

What the module is trying to prevent

This module is designed to generate, constrain, or falsify hypotheses. Its job is not to certify PRAYCG. It helps diagnose mechanisms, artifacts, timing issues, burden states, and alternative explanations.

Inputs and assumptions

Feature table, candidate events, theta/meaning/TSP windows.

The assumptions are narrow: the input files must correspond to the same run, use the same branch timing, and preserve the condition labels expected by the suite. If the module is anchor-based, anchor IDs and rendered times must match the final rendered stimulus rather than a remembered source-video time.

Algorithmic logic

Fit event-local reciprocal loop models using recognition features and delayed integration/topology features; compute K-HT-topo and percentiles.

In practice, the module reduces high-dimensional time series into branch-level, anchor-level, or cue-level summaries. The exact weights and thresholds are intended to be transparent and modifiable rather than hidden. For public replication, the lock-in file should freeze thresholds before analysis.

Outputs and where to find them

candidate_local_kht_topo_mred_event_table.csv, kht visual overlays.

These outputs usually live in the Master Suite output folder under tables/. Visualizer-specific sidecars are separate and should not be confused with canonical analysis tables.

How to interpret it

K-HT-topo is a human-scale coupling proxy. It is not K-OSM and not cellular proof.

The safest interpretation always asks three questions: Did Target exceed Control? Did Target exceed Override? Did artifact, timing, stimulus drive, or self-report contradict the interpretation?

How it can fail

High K without delayed theta/carryover or condition specificity is insufficient.

Safe public wording

Recommended wording
This module supports the analysis chain. It should be read as a bounded measurement or QC layer, not a standalone claim.

13. MRED - Meaning Recognition / Encoding Dissociation

Tier: Primary conceptual endpoint

Separates meaning recognition (MR) from encoding/integration (ENC).

What the module is trying to prevent

This module supports the core analysis chain. It should be interpreted only in relation to timing, artifact, Control, Override, and stimulus-regressor checks.

Inputs and assumptions

TSP, MeaningGamma, KHT-topo, theta handoff, topology/afterstate proxies, artifact and condition labels.

The assumptions are narrow: the input files must correspond to the same run, use the same branch timing, and preserve the condition labels expected by the suite. If the module is anchor-based, anchor IDs and rendered times must match the final rendered stimulus rather than a remembered source-video time.

Algorithmic logic

Compute MR and ENC scores at anchors/candidates. Classify quadrants such as MR_HIGH_ENC_HIGH or MR_HIGH_ENC_LOW.

In practice, the module reduces high-dimensional time series into branch-level, anchor-level, or cue-level summaries. The exact weights and thresholds are intended to be transparent and modifiable rather than hidden. For public replication, the lock-in file should freeze thresholds before analysis.

Outputs and where to find them

mred_event_table.csv, candidate_local_kht_topo_mred_event_table.csv.

These outputs usually live in the Master Suite output folder under tables/. Visualizer-specific sidecars are separate and should not be confused with canonical analysis tables.

How to interpret it

MRED is the conceptual core: recognition can appear without clear integration, and integration-like changes may be nonsemantic unless Target-specific.

The safest interpretation always asks three questions: Did Target exceed Control? Did Target exceed Override? Did artifact, timing, stimulus drive, or self-report contradict the interpretation?

How it can fail

No Control/Override specificity, high artifact, or no delayed window weakens classification.

Safe public wording

Recommended wording
This module supports the analysis chain. It should be read as a bounded measurement or QC layer, not a standalone claim.

14. A-MRED - Anchor-Locked MRED

Tier: Boxed primary path

Compresses the suite into a cleaner primary endpoint for future studies.

What the module is trying to prevent

This module supports the core analysis chain. It should be interpreted only in relation to timing, artifact, Control, Override, and stimulus-regressor checks.

Inputs and assumptions

Locked anchor JSON, feature table, MRED table, Control/Target/Override contrasts, QC gates, CET-R where available.

The assumptions are narrow: the input files must correspond to the same run, use the same branch timing, and preserve the condition labels expected by the suite. If the module is anchor-based, anchor IDs and rendered times must match the final rendered stimulus rather than a remembered source-video time.

Algorithmic logic

For each predeclared anchor, require Target MR and ENC above thresholds and greater than Control and Override, with QC pass.

In practice, the module reduces high-dimensional time series into branch-level, anchor-level, or cue-level summaries. The exact weights and thresholds are intended to be transparent and modifiable rather than hidden. For public replication, the lock-in file should freeze thresholds before analysis.

Outputs and where to find them

amred_anchor_endpoint_table.csv, amred_primary_endpoint_summary.csv, amred_visual_overlay.csv.

These outputs usually live in the Master Suite output folder under tables/. Visualizer-specific sidecars are separate and should not be confused with canonical analysis tables.

How to interpret it

A-MRED pass is the strongest current endpoint category. Strict use requires frame-verified locked anchors and clean timing.

The safest interpretation always asks three questions: Did Target exceed Control? Did Target exceed Override? Did artifact, timing, stimulus drive, or self-report contradict the interpretation?

How it can fail

Estimated anchors, weak ALS, missing Control/Override, or timing sensitivity produce pilot-positive, not confirmatory, results.

Safe public wording

Recommended wording

A predeclared anchor showed a Target-favoring candidate pattern under the specified gates. Strict confirmation still requires locked, frame-verified anchors, clean timing, artifact controls, and replication.

15. MRED-Peak

Tier: Boxed primary path / endpoint subtype

Identifies acute anchor-locked meaning shock or recognition-plus-integration events.

What the module is trying to prevent

This module supports the core analysis chain. It should be interpreted only in relation to timing, artifact, Control, Override, and stimulus-regressor checks.

Inputs and assumptions

A-MRED, BIT, MRED scores, CII margins, KHT/MR/ENC tables.

The assumptions are narrow: the input files must correspond to the same run, use the same branch timing, and preserve the condition labels expected by the suite. If the module is anchor-based, anchor IDs and rendered times must match the final rendered stimulus rather than a remembered source-video time.

Algorithmic logic

Combine target MR, target ENC, strict gates, CII margins, and QC flags into a peak score and gate label.

In practice, the module reduces high-dimensional time series into branch-level, anchor-level, or cue-level summaries. The exact weights and thresholds are intended to be transparent and modifiable rather than hidden. For public replication, the lock-in file should freeze thresholds before analysis.

Outputs and where to find them

mred_peak_resolution_anchor_table.csv, top_mred_peak_candidates.csv.

These outputs usually live in the Master Suite output folder under tables/. Visualizer-specific sidecars are separate and should not be confused with canonical analysis tables.

How to interpret it

Best for Contact-style acute recognition events. A strict peak pass still requires anchor/timing confidence.

The safest interpretation always asks three questions: Did Target exceed Control? Did Target exceed Override? Did artifact, timing, stimulus drive, or self-report contradict the interpretation?

How it can fail

Peak candidates without strict lock stay exploratory.

Safe public wording

Recommended wording

A predeclared anchor showed a Target-favoring candidate pattern under the specified gates. Strict confirmation still requires locked, frame-verified anchors, clean timing, artifact controls, and replication.

16. MRED-Resolution

Tier: Secondary/endpoint subtype

Identifies delayed reflective or regulatory recovery rather than a single acute peak.

What the module is trying to prevent

This module summarizes a pattern already estimated elsewhere. It is useful for comparing branches, runs, or anchor families, but it should not outrank the primary endpoint. It helps explain shape and direction, not final truth.

Inputs and assumptions

Baseline2 metrics, NUPI/RDI, EET, self-report echo, MRED/NIP scores.

The assumptions are narrow: the input files must correspond to the same run, use the same branch timing, and preserve the condition labels expected by the suite. If the module is anchor-based, anchor IDs and rendered times must match the final rendered stimulus rather than a remembered source-video time.

Algorithmic logic

Combine post-target recovery, TargetEchoedDuringBaseline2, EET afterstate echo, and RDI with anchor-level context.

In practice, the module reduces high-dimensional time series into branch-level, anchor-level, or cue-level summaries. The exact weights and thresholds are intended to be transparent and modifiable rather than hidden. For public replication, the lock-in file should freeze thresholds before analysis.

Outputs and where to find them

mred_peak_resolution_anchor_table.csv, top_mred_resolution_candidates.csv.

These outputs usually live in the Master Suite output folder under tables/. Visualizer-specific sidecars are separate and should not be confused with canonical analysis tables.

How to interpret it

Best for Field-of-Dreams-style closure/recovery. It is a profile, not proof of healing.

The safest interpretation always asks three questions: Did Target exceed Control? Did Target exceed Override? Did artifact, timing, stimulus drive, or self-report contradict the interpretation?

How it can fail

No Baseline2 means true resolution cannot be graded; proxy-only results are flagged.

Safe public wording

Recommended wording
The run showed a recovery- or resolution-like profile in the available proxies. This is not a clinical or thermodynamic proof; it is an interpretable after-state profile.

17. NIP - Narrative Immersion Proxy

Tier: Secondary

Translates attention + resonance into macroscopic PRAYCG proxies.

What the module is trying to prevent

This module summarizes a pattern already estimated elsewhere. It is useful for comparing branches, runs, or anchor families, but it should not outrank the primary endpoint. It helps explain shape and direction, not final truth.

Inputs and assumptions

MeaningGamma/TSP/NAST for semantic attention; theta/KHT/topology/API for integration; artifact/confounds.

The assumptions are narrow: the input files must correspond to the same run, use the same branch timing, and preserve the condition labels expected by the suite. If the module is anchor-based, anchor IDs and rendered times must match the final rendered stimulus rather than a remembered source-video time.

Algorithmic logic

Compute A_sem and R_int components; multiply/compose into immersion density.

In practice, the module reduces high-dimensional time series into branch-level, anchor-level, or cue-level summaries. The exact weights and thresholds are intended to be transparent and modifiable rather than hidden. For public replication, the lock-in file should freeze thresholds before analysis.

Outputs and where to find them

nip_component_timeseries.csv.

These outputs usually live in the Master Suite output folder under tables/. Visualizer-specific sidecars are separate and should not be confused with canonical analysis tables.

How to interpret it

NIP estimates attention-plus-integration immersion proxy, not dopamine/oxytocin or literal immersion chemistry.

The safest interpretation always asks three questions: Did Target exceed Control? Did Target exceed Override? Did artifact, timing, stimulus drive, or self-report contradict the interpretation?

How it can fail

High sensory drive or artifact can mimic attention; high recognition without integration should not be called full immersion.

Safe public wording

Recommended wording

This module supports the analysis chain. It should be read as a bounded measurement or QC layer, not a standalone claim.

18. BIT - Bivariate Immersion Threshold

Tier: Secondary

Ensures immersion is not a mere average of attention and resonance.

What the module is trying to prevent

This module exists primarily to prevent a false-positive interpretation. It asks whether the run, timing, stimulus, or signal quality is strong enough to allow any downstream claim. If this layer fails, the correct output is not a weaker positive result; it is a downgraded or non-interpretable run.

Inputs and assumptions

NIP components, MRED quadrants, artifact and condition gates.

The assumptions are narrow: the input files must correspond to the same run, use the same branch timing, and preserve the condition labels expected by the suite. If the module is anchor-based, anchor IDs and rendered times must match the final rendered stimulus rather than a remembered source-video time.

Algorithmic logic

Boolean gate requiring semantic attention and integration/resonance to exceed thresholds simultaneously, with condition specificity.

In practice, the module reduces high-dimensional time series into branch-level, anchor-level, or cue-level summaries. The exact weights and thresholds are intended to be transparent and modifiable rather than hidden. For public replication, the lock-in file should freeze thresholds before analysis.

Outputs and where to find them

bit_event_table.csv.

These outputs usually live in the Master Suite output folder under tables/. Visualizer-specific sidecars are separate and should not be confused with canonical analysis tables.

How to interpret it

BIT pass approximates a strict MR_HIGH_ENC_HIGH immersion candidate.

The safest interpretation always asks three questions: Did Target exceed Control? Did Target exceed Override? Did artifact, timing, stimulus drive, or self-report contradict the interpretation?

How it can fail

High attention but low integration fails BIT by design.

Safe public wording

Recommended wording
This module supports the analysis chain. It should be read as a bounded measurement or QC layer, not a standalone claim.

19. CII - Continuous Immersion Index

Tier: Secondary

Scores scene-level immersion density over anchor windows.

What the module is trying to prevent

This module summarizes a pattern already estimated elsewhere. It is useful for comparing branches, runs, or anchor families, but it should not outrank the primary endpoint. It helps explain shape and direction, not final truth.

Inputs and assumptions

NIP time series and anchor windows.

The assumptions are narrow: the input files must correspond to the same run, use the same branch timing, and preserve the condition labels expected by the suite. If the module is anchor-based, anchor IDs and rendered times must match the final rendered stimulus rather than a remembered source-video time.

Algorithmic logic

Integrate/average positive semantic-attention multiplied by integration proxy across a window, duration-normalized.

In practice, the module reduces high-dimensional time series into branch-level, anchor-level, or cue-level summaries. The exact weights and thresholds are intended to be transparent and modifiable rather than hidden. For public replication, the lock-in file should freeze thresholds before analysis.

Outputs and where to find them

cii_anchor_integrals.csv.

These outputs usually live in the Master Suite output folder under tables/. Visualizer-specific sidecars are separate and should not be confused with canonical analysis tables.

How to interpret it

Useful for sustained scenes, not just spikes. Longer windows must be normalized.

The safest interpretation always asks three questions: Did Target exceed Control? Did Target exceed Override? Did artifact, timing, stimulus drive, or self-report contradict the interpretation?

How it can fail

Can be inflated by duration, sensory rhythm, or artifacts if CET-R and QC are missing.

Safe public wording

Recommended wording
This module supports the analysis chain. It should be read as a bounded measurement or QC layer, not a standalone claim.

20. IAQ / ITQ - Immersion Attenuation

Tier: Secondary

Estimates how much Override attenuates Target immersion proxy.

What the module is trying to prevent

This module summarizes a pattern already estimated elsewhere. It is useful for comparing branches, runs, or anchor families, but it should not outrank the primary endpoint. It helps explain shape and direction, not final truth.

Inputs and assumptions

Target and Override CII/NIP values for matched anchors.

The assumptions are narrow: the input files must correspond to the same run, use the same branch timing, and preserve the condition labels expected by the suite. If the module is anchor-based, anchor IDs and rendered times must match the final rendered stimulus rather than a remembered source-video time.

Algorithmic logic

Compute Target-minus-Override difference or ratio-bounded attenuation.

In practice, the module reduces high-dimensional time series into branch-level, anchor-level, or cue-level summaries. The exact weights and thresholds are intended to be transparent and modifiable rather than hidden. For public replication, the lock-in file should freeze thresholds before analysis.

Outputs and where to find them

iaq_target_override_table.csv.

These outputs usually live in the Master Suite output folder under tables/. Visualizer-specific sidecars are separate and should not be confused with canonical analysis tables.

How to interpret it

Positive IAQ means Target preserved more immersion proxy than Override. It is not a moral theft score.

The safest interpretation always asks three questions: Did Target exceed Control? Did Target exceed Override? Did artifact, timing, stimulus drive, or self-report contradict the interpretation?

How it can fail

If Override equals/exceeds Target, reception/extraction claim weakens or story breaks through task.

Safe public wording

Recommended wording
This module supports the analysis chain. It should be read as a bounded measurement or QC layer, not a standalone claim.

21. TTI - Thermodynamic Theft Index

Tier: Secondary

Summarizes reception-vs-extraction tradeoff.

What the module is trying to prevent

This module summarizes a pattern already estimated elsewhere. It is useful for comparing branches, runs, or anchor families, but it should not outrank the primary endpoint. It helps explain shape and direction, not final truth.

Inputs and assumptions

MR, ENC, API-A/autonomic summaries, task/extraction metrics, artifact/visual penalties.

The assumptions are narrow: the input files must correspond to the same run, use the same branch timing, and preserve the condition labels expected by the suite. If the module is anchor-based, anchor IDs and rendered times must match the final rendered stimulus rather than a remembered source-video time.

Algorithmic logic

Weighted contrast: Target meaning/integration/regulation minus Override extraction/task load, penalized for artifacts and visual differences.

In practice, the module reduces high-dimensional time series into branch-level, anchor-level, or cue-level summaries. The exact weights and thresholds are intended to be transparent and modifiable rather than hidden. For public replication, the lock-in file should freeze thresholds before analysis.

Outputs and where to find them

tti_global_summary.csv, tti_anchor_deltas.csv, tti_visual_overlay.csv.

These outputs usually live in the Master Suite output folder under tables/. Visualizer-specific sidecars are separate and should not be confused with canonical analysis tables.

How to interpret it

Positive TTI means Target preserved more receptive meaning/integration while Override carried more extractive load.

The safest interpretation always asks three questions: Did Target exceed Control? Did Target exceed Override? Did artifact, timing, stimulus drive, or self-report contradict the interpretation?

How it can fail

Not literal thermodynamics, morality, or clinical metric. Artifact or task confounds can dominate.

Safe public wording

Recommended wording
This module supports the analysis chain. It should be read as a bounded measurement or QC layer, not a standalone claim.

22. NUPI - Narrative Update Polarity Index

Tier: Secondary

Classifies update profile as accommodative load versus resolute recovery.

What the module is trying to prevent

This module summarizes a pattern already estimated elsewhere. It is useful for comparing branches, runs, or anchor families, but it should not outrank the primary endpoint. It helps explain shape and direction, not final truth.

Inputs and assumptions

ALI components, RDI components, Baseline2, NUPI anchors, self-report echo, EET.

The assumptions are narrow: the input files must correspond to the same run, use the same branch timing, and preserve the condition labels expected by the suite. If the module is anchor-based, anchor IDs and rendered times must match the final rendered stimulus rather than a remembered source-video time.

Algorithmic logic

Compute ALI as semantic/update-load profile; compute RDI as regulatory-dividend/afterstate profile; NUPI = RDI - ALI.

In practice, the module reduces high-dimensional time series into branch-level, anchor-level, or cue-level summaries. The exact weights and thresholds are intended to be transparent and modifiable rather than hidden. For public replication, the lock-in file should freeze thresholds before analysis.

Outputs and where to find them

nupi_run_summary.csv, nupi_anchor_polarity_table.csv, nupi_visual_overlay.csv.

These outputs usually live in the Master Suite output folder under tables/. Visualizer-specific sidecars are separate and should not be confused with canonical analysis tables.

How to interpret it

Distinguishes high-load-with-recovery from resolute-recovery profiles. Does not measure heat/ATP/clinical healing.

The safest interpretation always asks three questions: Did Target exceed Control? Did Target exceed Override? Did artifact, timing, stimulus drive, or self-report contradict the interpretation?

How it can fail

No Baseline2 or no afterstate physiology prevents reliable polarity grading.

Safe public wording

Recommended wording
The run showed a recovery- or resolution-like profile in the available proxies. This is not a clinical or thermodynamic proof; it is an interpretable after-state profile.

23. NAST - Narrative Absorption State Transition

Tier: Exploratory / transition

Detects branch-entry state transition into narrative absorption or task stance.

What the module is trying to prevent

This module is designed to generate, constrain, or falsify hypotheses. Its job is not to certify PRAYCG. It helps diagnose mechanisms, artifacts, timing issues, burden states, and alternative explanations.

Inputs and assumptions

Late washout windows, early branch windows, alpha, MeaningGamma, TSP, TaskGamma, theta.

The assumptions are narrow: the input files must correspond to the same run, use the same branch timing, and preserve the condition labels expected by the suite. If the module is anchor-based, anchor IDs and rendered times must match the final rendered stimulus rather than a remembered source-video time.

Algorithmic logic

Compare pre-branch and early-branch features; quantify meaning/alpha/task transitions.

In practice, the module reduces high-dimensional time series into branch-level, anchor-level, or cue-level summaries. The exact weights and thresholds are intended to be transparent and modifiable rather than hidden. For public replication, the lock-in file should freeze thresholds before analysis.

Outputs and where to find them

nast_phase_transition_table.csv.

These outputs usually live in the Master Suite output folder under tables/. Visualizer-specific sidecars are separate and should not be confused with canonical analysis tables.

How to interpret it

Useful for onset dynamics. Target may show meaning/TSP increase; Override may show task-gamma increase.

The safest interpretation always asks three questions: Did Target exceed Control? Did Target exceed Override? Did artifact, timing, stimulus drive, or self-report contradict the interpretation?

How it can fail

Alpha is eye/attention sensitive; NAST alone is not a meaning endpoint.

Safe public wording

Recommended wording
This module supports the analysis chain. It should be read as a bounded measurement or QC layer, not a standalone claim.

24. Topo-OSM NetworkState

Tier: Exploratory interpretation

Represents human-scale latent state as network-state topology proxy.

What the module is trying to prevent

This module is designed to generate, constrain, or falsify hypotheses. Its job is not to certify PRAYCG. It helps diagnose mechanisms, artifacts, timing issues, burden states, and alternative explanations.

Inputs and assumptions

Theta, gamma, alpha, TSP, API-A, respiration, artifact, branch summaries.

The assumptions are narrow: the input files must correspond to the same run, use the same branch timing, and preserve the condition labels expected by the suite. If the module is anchor-based, anchor IDs and rendered times must match the final rendered stimulus rather than a remembered source-video time.

Algorithmic logic

Build phase/branch vectors and transition distances rather than cellular hidden-Y claims.

In practice, the module reduces high-dimensional time series into branch-level, anchor-level, or cue-level summaries. The exact weights and thresholds are intended to be transparent and modifiable rather than hidden. For public replication, the lock-in file should freeze thresholds before analysis.

Outputs and where to find them

topo_osm_phase_state_vectors.csv, topo_osm_visual_overlay.csv.

These outputs usually live in the Master Suite output folder under tables/. Visualizer-specific sidecars are separate and should not be confused with canonical analysis tables.

How to interpret it

Human-scale Topo-OSM only: state geometry, not microtubules or biophotons.

The safest interpretation always asks three questions: Did Target exceed Control? Did Target exceed Override? Did artifact, timing, stimulus drive, or self-report contradict the interpretation?

How it can fail

Too few channels, unconfirmed channel map, or artifact can make topology labels unreliable.

Safe public wording

Recommended wording

This module supports the analysis chain. It should be read as a bounded measurement or QC layer, not a standalone claim.

25. EET - Endogenous Echo Tracking

Tier: Exploratory convergence

Tests whether scene-anchor state vectors resemble later washout/Baseline2 states.

What the module is trying to prevent

This module is designed to generate, constrain, or falsify hypotheses. Its job is not to certify PRAYCG. It helps diagnose mechanisms, artifacts, timing issues, burden states, and alternative explanations.

Inputs and assumptions

Anchor feature vectors, washout/Baseline2 feature vectors, NIP/MRED/EET tables.

The assumptions are narrow: the input files must correspond to the same run, use the same branch timing, and preserve the condition labels expected by the suite. If the module is anchor-based, anchor IDs and rendered times must match the final rendered stimulus rather than a remembered source-video time.

Algorithmic logic

Compute cosine/similarity or vector resemblance between anchor state and later afterstate windows.

In practice, the module reduces high-dimensional time series into branch-level, anchor-level, or cue-level summaries. The exact weights and thresholds are intended to be transparent and modifiable rather than hidden. For public replication, the lock-in file should freeze thresholds before analysis.

Outputs and where to find them

eet_endogenous_echo_tracking.csv.

These outputs usually live in the Master Suite output folder under tables/. Visualizer-specific sidecars are separate and should not be confused with canonical analysis tables.

How to interpret it

Suggests afterstate resemblance. It is not replay proof or memory proof.

The safest interpretation always asks three questions: Did Target exceed Control? Did Target exceed Override? Did artifact, timing, stimulus drive, or self-report contradict the interpretation?

How it can fail

Similarity may reflect baseline drift, common arousal, artifact, or generic state persistence.

Safe public wording

Recommended wording

The after-state resembled the anchor-state vector under the specified metric. This is not proof of replay or memory.

26. MRED-ITP - Information-Thermodynamic Proxy

Tier: Exploratory convergence

Adds complexity and ocular-release proxies around MRED events.

What the module is trying to prevent

This module is designed to generate, constrain, or falsify hypotheses. Its job is not to certify PRAYCG. It helps diagnose mechanisms, artifacts, timing issues, burden states, and alternative explanations.

Inputs and assumptions

LZ-style complexity time series, blink/Fp1 proxy, MRED anchors, artifact.

The assumptions are narrow: the input files must correspond to the same run, use the same branch timing, and preserve the condition labels expected by the suite. If the module is anchor-based, anchor IDs and rendered times must match the final rendered stimulus rather than a remembered source-video time.

Algorithmic logic

Compute complexity strike/settlement and blink suppression/release metrics, then summarize convergence with MRED.

In practice, the module reduces high-dimensional time series into branch-level, anchor-level, or cue-level summaries. The exact weights and thresholds are intended to be transparent and modifiable rather than hidden. For public replication, the lock-in file should freeze thresholds before analysis.

Outputs and where to find them

lz_complexity_timeseries.csv, acg_event_table.csv, blink_event_table.csv, ocu_event_table.csv, mred_itp_anchor_summary.csv.

These outputs usually live in the Master Suite output folder under tables/. Visualizer-specific sidecars are separate and should not be confused with canonical analysis tables.

How to interpret it

A convergence layer only. Complexity is not thermodynamic entropy; blink release is not memory encoding proof.

The safest interpretation always asks three questions: Did Target exceed Control? Did Target exceed Override? Did artifact, timing, stimulus drive, or self-report contradict the interpretation?

How it can fail

High artifact, lack of EOG/eye tracking, or modality-specific complexity effects limit interpretation.

Safe public wording

Recommended wording

This module supplied convergence or falsification evidence. It does not prove entropy reduction or memory encoding.

27. ACG - Algorithmic Complexity Gate

Tier: Exploratory

Looks for complexity perturbation and settlement around anchors.

What the module is trying to prevent

This module is designed to generate, constrain, or falsify hypotheses. Its job is not to certify PRAYCG. It helps diagnose mechanisms, artifacts, timing issues, burden states, and alternative explanations.

Inputs and assumptions

Feature-state or raw EEG symbolic complexity windows.

The assumptions are narrow: the input files must correspond to the same run, use the same branch timing, and preserve the condition labels expected by the suite. If the module is anchor-based, anchor IDs and rendered times must match the final rendered stimulus rather than a remembered source-video time.

Algorithmic logic

Compare pre, peak, and post LZ-style complexity: strike = peak-pre; settle = post-peak.

In practice, the module reduces high-dimensional time series into branch-level, anchor-level, or cue-level summaries. The exact weights and thresholds are intended to be transparent and modifiable rather than hidden. For public replication, the lock-in file should freeze thresholds before analysis.

Outputs and where to find them

acg_event_table.csv, lz_complexity_timeseries.csv.

These outputs usually live in the Master Suite output folder under tables/. Visualizer-specific sidecars are separate and should not be confused with canonical analysis tables.

How to interpret it

Structured complexity trajectory can support perturbation/settlement interpretation.

The safest interpretation always asks three questions: Did Target exceed Control? Did Target exceed Override? Did artifact, timing, stimulus drive, or self-report contradict the interpretation?

How it can fail

Not literal entropy; artifact/noise can raise complexity; drowsiness/flatline can lower it.

Safe public wording

Recommended wording

This module supplied convergence or falsification evidence. It does not prove entropy reduction or memory encoding.

28. OCU - Ocular-Cognitive Unloading

Tier: Exploratory / ocular

Tracks blink suppression and release around narrative anchors.

What the module is trying to prevent

This module is designed to generate, constrain, or falsify hypotheses. Its job is not to certify PRAYCG. It helps diagnose mechanisms, artifacts, timing issues, burden states, and alternative explanations.

Inputs and assumptions

Fp1/frontal ocular proxy or EOG if available, anchor windows, artifact table.

The assumptions are narrow: the input files must correspond to the same run, use the same branch timing, and preserve the condition labels expected by the suite. If the module is anchor-based, anchor IDs and rendered times must match the final rendered stimulus rather than a remembered source-video time.

Algorithmic logic

Detect blink-like events, compute blink rate before/during/after anchor, quantify suppression/release.

In practice, the module reduces high-dimensional time series into branch-level, anchor-level, or cue-level summaries. The exact weights and thresholds are intended to be transparent and modifiable rather than hidden. For public replication, the lock-in file should freeze thresholds before analysis.

Outputs and where to find them

blink_event_table.csv, ocu_event_table.csv.

These outputs usually live in the Master Suite output folder under tables/. Visualizer-specific sidecars are separate and should not be confused with canonical analysis tables.

How to interpret it

Possible event-boundary/attention-release marker. Stronger with EOG/eye tracking.

The safest interpretation always asks three questions: Did Target exceed Control? Did Target exceed Override? Did artifact, timing, stimulus drive, or self-report contradict the interpretation?

How it can fail

Fp1 proxy confounds blink with squint, EMG, eye strain, fatigue, or electrode motion.

Safe public wording

Recommended wording

This module supplied convergence or falsification evidence. It does not prove entropy reduction or memory encoding.

29. OCM025 - Override Cue Microstate

Tier: Override diagnostic

Analyzes digit recognition and working-memory updates around number cues at 0.25 s resolution.

What the module is trying to prevent

This module is designed to generate, constrain, or falsify hypotheses. Its job is not to certify PRAYCG. It helps diagnose mechanisms, artifacts, timing issues, burden states, and alternative explanations.

Inputs and assumptions

Cue schedule, raw/feature windows, Override/Target/Control comparisons.

The assumptions are narrow: the input files must correspond to the same run, use the same branch timing, and preserve the condition labels expected by the suite. If the module is anchor-based, anchor IDs and rendered times must match the final rendered stimulus rather than a remembered source-video time.

Algorithmic logic

Segment cue epochs into pre-cue, digit recognition, working-memory update, and maintenance windows; compare to nulls.

In practice, the module reduces high-dimensional time series into branch-level, anchor-level, or cue-level summaries. The exact weights and thresholds are intended to be transparent and modifiable rather than hidden. For public replication, the lock-in file should freeze thresholds before analysis.

Outputs and where to find them

ocm025_rsm_cvb_squint_cue_epoch_table.csv.

These outputs usually live in the Master Suite output folder under tables/. Visualizer-specific sidecars are separate and should not be confused with canonical analysis tables.

How to interpret it

Tests whether Override task was physiologically engaged. Not a meaning endpoint.

The safest interpretation always asks three questions: Did Target exceed Control? Did Target exceed Override? Did artifact, timing, stimulus drive, or self-report contradict the interpretation?

How it can fail

Poor cue visibility, missed cues, or artifacts can dominate.

Safe public wording

Recommended wording
This module describes task, cue, or visual-burden structure in Override. It does not prove a private mental strategy.

30. RSM - Running Sum Microstate Model

Tier: Override diagnostic

Models arithmetic load, compute stall, deadline pressure, and latent guess risk.

What the module is trying to prevent

This module is designed to generate, constrain, or falsify hypotheses. Its job is not to certify PRAYCG. It helps diagnose mechanisms, artifacts, timing issues, burden states, and alternative explanations.

Inputs and assumptions

Cue values, running-sum trajectory, OCM025 features, task self-report.

The assumptions are narrow: the input files must correspond to the same run, use the same branch timing, and preserve the condition labels expected by the suite. If the module is anchor-based, anchor IDs and rendered times must match the final rendered stimulus rather than a remembered source-video time.

Algorithmic logic

Compute carry difficulty, high-carry load, hard-9 patterns, theta maintenance, closure delay, and latent guess-risk proxies.

In practice, the module reduces high-dimensional time series into branch-level, anchor-level, or cue-level summaries. The exact weights and thresholds are intended to be transparent and modifiable rather than hidden. For public replication, the lock-in file should freeze thresholds before analysis.

Outputs and where to find them

rsm_correlation_summary.csv, cue tables.

These outputs usually live in the Master Suite output folder under tables/. Visualizer-specific sidecars are separate and should not be confused with canonical analysis tables.

How to interpret it

Identifies candidate burden/strategy windows, not proof of private strategy.

The safest interpretation always asks three questions: Did Target exceed Control? Did Target exceed Override? Did artifact, timing, stimulus drive, or self-report contradict the interpretation?

How it can fail

Requires accurate cue schedule and task report; cannot prove actual guessing.

Safe public wording

Recommended wording
This module describes task, cue, or visual-burden structure in Override. It does not prove a private mental strategy.

31. CVB - Cue Visibility / Legibility Burden

Tier: Override diagnostic

Separates cue perception burden from arithmetic compute burden.

What the module is trying to prevent

This module is designed to generate, constrain, or falsify hypotheses. Its job is not to certify PRAYCG. It helps diagnose mechanisms, artifacts, timing issues, burden states, and alternative explanations.

Inputs and assumptions

Cue schedule, visual contrast/legibility metrics, OCM/RSM features, SquintProxy.

The assumptions are narrow: the input files must correspond to the same run, use the same branch timing, and preserve the condition labels expected by the suite. If the module is anchor-based, anchor IDs and rendered times must match the final rendered stimulus rather than a remembered source-video time.

Algorithmic logic

Combine weak digit-recognition, high holding burden, low contrast/blur reports, and visual-search proxies.

In practice, the module reduces high-dimensional time series into branch-level, anchor-level, or cue-level summaries. The exact weights and thresholds are intended to be transparent and modifiable rather than hidden. For public replication, the lock-in file should freeze thresholds before analysis.

Outputs and where to find them

cue legibility burden columns and candidate tables.

These outputs usually live in the Master Suite output folder under tables/. Visualizer-specific sidecars are separate and should not be confused with canonical analysis tables.

How to interpret it

Flags possible hard-to-see cues. Particularly useful for Snack Attack and Her-like issues.

The safest interpretation always asks three questions: Did Target exceed Control? Did Target exceed Override? Did artifact, timing, stimulus drive, or self-report contradict the interpretation?

How it can fail

Cannot confirm invisibility without frame-level visibility and/or eye tracking.

Safe public wording

Recommended wording
This module describes task, cue, or visual-burden structure in Override. It does not prove a private mental strategy.

32. SquintProxy

Tier: Artifact/covariate

Uses frontal/Fp1-like artifact as possible visual strain sentinel.

What the module is trying to prevent

This module supports the core analysis chain. It should be interpreted only in relation to timing, artifact, Control, Override, and stimulus-regressor checks.

Inputs and assumptions

Fp1/frontal high-frequency, p2p, blink-like sentinels, self-report.

The assumptions are narrow: the input files must correspond to the same run, use the same branch timing, and preserve the condition labels expected by the suite. If the module is anchor-based, anchor IDs and rendered times must match the final rendered stimulus rather than a remembered source-video time.

Algorithmic logic

Score eye-strain/squint candidates separately from blink events and EEG endpoints.

In practice, the module reduces high-dimensional time series into branch-level, anchor-level, or cue-level summaries. The exact weights and thresholds are intended to be transparent and modifiable rather than hidden. For public replication, the lock-in file should freeze thresholds before analysis.

Outputs and where to find them

squint_proxy candidates within RSM/CVB tables.

These outputs usually live in the Master Suite output folder under tables/. Visualizer-specific sidecars are separate and should not be confused with canonical analysis tables.

How to interpret it

Useful as covariate only. EOG/eye tracking is needed for confirmation.

The safest interpretation always asks three questions: Did Target exceed Control? Did Target exceed Override? Did artifact, timing, stimulus drive, or self-report contradict the interpretation?

How it can fail

Confounds with blink, EMG, brow tension, electrode movement, general effort.

Safe public wording

Recommended wording
This module describes task, cue, or visual-burden structure in Override. It does not prove a private mental strategy.

33. API-A / Autonomic availability

Tier: Secondary/covariate

Summarizes regulated availability rather than simple arousal.

What the module is trying to prevent

This module summarizes a pattern already estimated elsewhere. It is useful for comparing branches, runs, or anchor families, but it should not outrank the primary endpoint. It helps explain shape and direction, not final truth.

Inputs and assumptions

HR, RMSSD/SDNN/pNN50, HR slope, respiration if available.

The assumptions are narrow: the input files must correspond to the same run, use the same branch timing, and preserve the condition labels expected by the suite. If the module is anchor-based, anchor IDs and rendered times must match the final rendered stimulus rather than a remembered source-video time.

Algorithmic logic

Weighted z-score composite favoring lower HR, higher HRV, stable/regulated trends, and respiratory context.

In practice, the module reduces high-dimensional time series into branch-level, anchor-level, or cue-level summaries. The exact weights and thresholds are intended to be transparent and modifiable rather than hidden. For public replication, the lock-in file should freeze thresholds before analysis.

Outputs and where to find them

api_a columns, autonomic summaries.

These outputs usually live in the Master Suite output folder under tables/. Visualizer-specific sidecars are separate and should not be confused with canonical analysis tables.

How to interpret it

Helps distinguish receptive regulation from strain, but not a clinical measure.

The safest interpretation always asks three questions: Did Target exceed Control? Did Target exceed Override? Did artifact, timing, stimulus drive, or self-report contradict the interpretation?

How it can fail

Requires reliable R-R/ECG; respiration/common drive can explain HRV.

Safe public wording

Recommended wording
This module supports the analysis chain. It should be read as a bounded measurement or QC layer, not a standalone claim.

34. Baseline 1 vs Baseline 2

Tier: Secondary afterstate

Compares pre-run baseline to final reflection baseline.

What the module is trying to prevent

This module summarizes a pattern already estimated elsewhere. It is useful for comparing branches, runs, or anchor families, but it should not outrank the primary endpoint. It helps explain shape and direction, not final truth.

Inputs and assumptions

Baseline phase features, HR/HRV, respiration, NIP/TSP/TaskGamma, artifact.

The assumptions are narrow: the input files must correspond to the same run, use the same branch timing, and preserve the condition labels expected by the suite. If the module is anchor-based, anchor IDs and rendered times must match the final rendered stimulus rather than a remembered source-video time.

Algorithmic logic

Compute feature deltas and regulation/reflective-state indicators.

In practice, the module reduces high-dimensional time series into branch-level, anchor-level, or cue-level summaries. The exact weights and thresholds are intended to be transparent and modifiable rather than hidden. For public replication, the lock-in file should freeze thresholds before analysis.

Outputs and where to find them

baseline1_vs_baseline2_summary.csv.

These outputs usually live in the Master Suite output folder under tables/. Visualizer-specific sidecars are separate and should not be confused with canonical analysis tables.

How to interpret it

Supports afterglow/recovery interpretation when Baseline2 is cleaner, more regulated, or semantically active.

The safest interpretation always asks three questions: Did Target exceed Control? Did Target exceed Override? Did artifact, timing, stimulus drive, or self-report contradict the interpretation?

How it can fail

Can be contaminated by fatigue, report burden, order effects, or artifact differences.

Safe public wording

Recommended wording
This module supports the analysis chain. It should be read as a bounded measurement or QC layer, not a standalone claim.

35. LSO/SPM Subtitle Override

Tier: Optional exploratory

Analyzes subtitle-only or subtitle-mediated override variants.

What the module is trying to prevent

This module is designed to generate, constrain, or falsify hypotheses. Its job is not to certify PRAYCG. It helps diagnose mechanisms, artifacts, timing issues, burden states, and alternative explanations.

Inputs and assumptions

Subtitle file, bounding boxes, optional gaze, feature table, MRED/NIP outputs.

The assumptions are narrow: the input files must correspond to the same run, use the same branch timing, and preserve the condition labels expected by the suite. If the module is anchor-based, anchor IDs and rendered times must match the final rendered stimulus rather than a remembered source-video time.

Algorithmic logic

Compute lexical extraction cost, subtitle phase mapping, semantic timing shift, and subtitle MRED patterns.

In practice, the module reduces high-dimensional time series into branch-level, anchor-level, or cue-level summaries. The exact weights and thresholds are intended to be transparent and modifiable rather than hidden. For public replication, the lock-in file should freeze thresholds before analysis.

Outputs and where to find them

subtitle_line_event_table.csv, subtitle_phase_shift_table.csv, lso_visual_overlay.csv.

These outputs usually live in the Master Suite output folder under tables/. Visualizer-specific sidecars are separate and should not be confused with canonical analysis tables.

How to interpret it

Tests route cost: semantic access through text versus natural audiovisual absorption.

The safest interpretation always asks three questions: Did Target exceed Control? Did Target exceed Override? Did artifact, timing, stimulus drive, or self-report contradict the interpretation?

How it can fail

Without gaze, reading timing is estimated only. Audio+subtitle variants test a different condition.

Safe public wording

Recommended wording
This module supports the analysis chain. It should be read as a bounded measurement or QC layer, not a standalone claim.

36. MasterSync Visualizer

Tier: Visualization/audit

Creates synchronized stimulus/data review videos with public/TSC or lab-diagnostic modes.

What the module is trying to prevent

This module supports the core analysis chain. It should be interpreted only in relation to timing, artifact, Control, Override, and stimulus-regressor checks.

Inputs and assumptions

Stimulus MP4, XDF or feature CSV, analysis output folder, overlay tables.

The assumptions are narrow: the input files must correspond to the same run, use the same branch timing, and preserve the condition labels expected by the suite. If the module is anchor-based, anchor IDs and rendered times must match the final rendered stimulus rather than a remembered source-video time.

Algorithmic logic

Slice branch by markers, repair timebases when possible, robust-scale panels, show missing-data labels, render event overlays/cards.

In practice, the module reduces high-dimensional time series into branch-level, anchor-level, or cue-level summaries. The exact weights and thresholds are intended to be transparent and modifiable rather than hidden. For public replication, the lock-in file should freeze thresholds before analysis.

Outputs and where to find them

analysis MP4, features_used.csv, events_used.csv, render_report.json.

These outputs usually live in the Master Suite output folder under tables/. Visualizer-specific sidecars are separate and should not be confused with canonical analysis tables.

How to interpret it

Excellent for review and explanation. Does not certify endpoint validity.

The safest interpretation always asks three questions: Did Target exceed Control? Did Target exceed Override? Did artifact, timing, stimulus drive, or self-report contradict the interpretation?

How it can fail

Bad feature table, missing panels, or uncorrected event timebase can mislead.

Safe public wording

Recommended wording
This module supports the analysis chain. It should be read as a bounded measurement or QC layer, not a standalone claim.

37. Offline Interpretive Report Generator

Tier: Reporting / usability

Generates a local rule-based plain-English report from a Master Suite output folder.

What the module is trying to prevent

This module supports the core analysis chain. It should be interpreted only in relation to timing, artifact, Control, Override, and stimulus-regressor checks.

Inputs and assumptions

Analysis folder tables, optional auto-run MRED-Peak/Resolution.

The assumptions are narrow: the input files must correspond to the same run, use the same branch timing, and preserve the condition labels expected by the suite. If the module is anchor-based, anchor IDs and rendered times must match the final rendered stimulus rather than a remembered source-video time.

Algorithmic logic

Detect available tables, summarize QC, A-MRED, MRED-Peak/Resolution, NIP/TTI/NUPI/CET/EET/ITP/OCM/RSM, and boundaries.

In practice, the module reduces high-dimensional time series into branch-level, anchor-level, or cue-level summaries. The exact weights and thresholds are intended to be transparent and modifiable rather than hidden. For public replication, the lock-in file should freeze thresholds before analysis.

Outputs and where to find them

offline_interpretive_report.md/txt/json.

These outputs usually live in the Master Suite output folder under tables/. Visualizer-specific sidecars are separate and should not be confused with canonical analysis tables.

How to interpret it

Helps GitHub/OSF users interpret outputs without uploading data to an AI assistant.

The safest interpretation always asks three questions: Did Target exceed Control? Did Target exceed Override? Did artifact, timing, stimulus drive, or self-report contradict the interpretation?

How it can fail

Rule-based, not flexible scientific reasoning. It cannot rescue bad inputs or discover unmodeled confounds.

Safe public wording

Recommended wording
This module supports the analysis chain. It should be read as a bounded measurement or QC layer, not a standalone claim.

Final recommended interpretation workflow

- Start with run identity, media hashes, and Target/Override hash match.
- Check channel map, stream inventory, phase coverage, and ALS/display timing.
- Check artifact and confound tables before looking at gamma or theta.
- Check StimulusFingerprint/CET-R to see whether movie rhythm or cue rhythm explains the pattern.
- Use A-MRED/MRED-Peak/Resolution as the primary interpretive compression.

- Use NIP/TTI/NUPI as secondary summaries of immersion, extraction, and polarity.
- Use KHT-topo, NAST, EET, MRED-ITP, OCM/RSM/CVB/Squint as convergence or falsification layers.
- Read self-report as context and a consistency check, not proof.
- If the offline interpreter gives a positive summary but the QC layer failed, the QC layer wins.

Glossary of claim grades

Grade	Meaning
Strict confirmatory candidate	Locked, frame-verified anchors; clean timing; artifact/CET-R pass; Target > Control and Target > Override; preregistered rule.
Runner-registered estimated pilot	Anchor file loaded before acquisition, but times are estimated or not frame-verified. Useful but not confirmatory.
Conceptually predeclared exploratory	Anchor family existed before analysis, but was not loaded into the runner. Stronger than pure fishing, weaker than formal registration.
Exploratory discovery	Found after analysis. Useful for future-freezing only.
Cautioned / non-interpretable	Timing, artifact, stimulus, channel-map, or confound problem prevents strong interpretation.